

FINAL PROJECT ON DATA ANALYTICS

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TITLE: Human Resource Management and Employee Data Analysis

PROJECT DOMAIN: Human Resource

TIMELINE: 2010 to 2023

SUBMISSION DATE: 01.04.2026

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DATA SET LINK:

Raw Data Set:

https://github.com/Dhivya-S92/mydataset/blob/1bb46a5125adebcda89070433732cd11c24a8d9a/HR_Uncleaned_Dataset.xlsx

Cleaned data set:

https://docs.google.com/spreadsheets/d/1PFghLU0A879Hq4cNg2QCEVtZ3PWJ1uq3/edit?usp=drive_link&ouid=111532347864707315771&rtpof=true&sd=true

Human Resource Management and Employee Data Analysis

OBJECTIVE

1. To manage Employee Information such as Personal information (Name, Gender, Date of Birth), Contact details (Email, Phone), Job-related data (Department, Job Title, Location).
2. To support HR Decision-Making and analyze using Salary and bonus data, Performance ratings and Experience and education levels.
3. To track Employee Performance & Development through Performance ratings, Training hours and Promotions and experience.
4. To monitor Workforce Status using Active/inactive employee status, Leaves taken and Manager relationships.
5. To enable Data Analysis & Reporting by Provide a dataset for HR analytics, Reporting dashboards and Predictive modeling.
6. To clean the data such as Handling missing values, Fixing inconsistent formats and Preparing data for analysis.
7. To Meaningful visualizations such as bar charts, pie charts, histograms, and scatter plots help reveal patterns in employee distribution, salary structure, performance, and training effectiveness, making the data easier to understand and interpret.
8. To provides insights into employee distribution, performance, salary structure, training effectiveness, and overall workforce trends, enabling data-driven HR decisions.
9. To provide actionable insights and data-driven recommendations based on the analysis to support effective decision-making.

OUTCOME

The database enables efficient management, analysis, and visualization of employee data, leading to actionable insights and improved HR decision-making.

Human Resource Management and Employee Data Analysis

DATASET INFORMATION

Source: Company Internal Dataset

Year / Timeline: Data collected during 2010 to 2023

Dataset Description:

1. Employee_ID – Unique identifier for each employee
2. Name – Employee name
3. Age – Age of the employee
4. Gender – Gender of the employee
5. Department – Department in which the employee works
6. Job_Title – Role/designation of the employee
7. Salary – Employee salary
8. Experience_Years – Years of work experience
9. Education_Level – Qualification of the employee
10. Performance_Rating – Employee performance score
11. Training_Hours – Number of training hours completed
12. Location – Work location
13. Joining_Date – Date of joining
14. Manager_ID – ID of the reporting manager

TYPE OF ANALYSIS

Descriptive Analysis (summarizing the dataset):

Summarizes employee data using averages, counts, and percentages

Examples: average salary, employee count by department

Human Resource Management and Employee Data Analysis

STAGE 1 – Problem Definition and Dataset Selection

Define the business problem and expected outcome

Choose dataset and explain its source, size, and features

Import libraries, load dataset

Dataset description (rows, columns, features)

Initial EDA (head, info, describe, shape, null checks, duplicate check)

```
# Import required libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from google.colab import files
uploaded = files.upload()
```

Choose files No file chosen

Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

Saving HR_Uncleaned_Dataset.xlsx to HR_Uncleaned_Dataset.xlsx

```
import pandas as pd

df = pd.read_excel("HR_Uncleaned_Dataset.xlsx")
df.head()
```

Employee_ID	First_Name	Last_Name	Gender	Date_of_Birth	Email	Phone	Department	Job_Title	Location	Date_of_Joining	Salary	Performance_Rating	Years_of_Experience
0	EMP2886	Sneha	Wang	Male	1995-12-01	NaN	NaN	Finance	VP	Mumbai	2016-04-02	67859.83	2.0
1	EMP2058	Yuki	Ahmed	Female	1984-10-09	deepa.brown68@company.com	+91-6624030165	Admin	VP	Dubai	2011-07-15	77907.72	4.0
2	EMP5453	James	Ahmed	Other	06-24-1973	layla.wang86@company.com	+91-6987938413	IT	Associate	Singapore	2023-07-07	160139.77	5.0
3	EMP2412	Deepa	Johnson	Male	1998-12-18	hiroshi.sharma54@company.com	+91-9379729728	Legal	Associate	Singapore	2016-12-27	30176.64	1.0
4	EMP2128	Fatima	Jones	Female	1977-05-07	NaN	+91-8303574566	R&D	Coordinator	Sydney	2011-02-06	126357.31	2.0

```
df.shape
```

```
(5080, 20)
```

```
df.columns
```

```
... Index(['Employee_ID', 'First_Name', 'Last_Name', 'Gender', 'Date_of_Birth',
          'Email', 'Phone', 'Department', 'Job_Title', 'Location',
          'Date_of_Joining', 'Salary', 'Performance_Rating',
          'Years_of_Experience', 'Education_Level', 'Is_Active',
          'Bonus_Percentage', 'Training_Hours', 'Leaves_Taken', 'Manager_ID'],
          dtype='object')
```

Human Resource Management and Employee Data Analysis

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5080 entries, 0 to 5079
Data columns (total 20 columns):
#   Column              Non-Null Count  Dtype  
---  -
0   Employee_ID         5080 non-null   object  
1   First_Name          5080 non-null   object  
2   Last_Name           5080 non-null   object  
3   Gender              4928 non-null   object  
4   Date_of_Birth       4879 non-null   object  
5   Email               4878 non-null   object  
6   Phone               4774 non-null   object  
7   Department          5080 non-null   object  
8   Job_Title           5080 non-null   object  
9   Location            5080 non-null   object  
10  Date_of_Joining     5080 non-null   object  
11  Salary              4829 non-null   float64 
12  Performance_Rating  4726 non-null   float64 
13  Years_of_Experience 5080 non-null   int64   
14  Education_Level     4674 non-null   object  
15  Is_Active           5080 non-null   object  
16  Bonus_Percentage    4829 non-null   float64 
17  Training_Hours      4780 non-null   float64 
18  Leaves_Taken        5080 non-null   int64   
19  Manager_ID          4877 non-null   object  
dtypes: float64(4), int64(2), object(14)
```

```
df.describe()
```

	Salary	Performance_Rating	Years_of_Experience	Bonus_Percentage	Training_Hours	Leaves_Taken
count	4.829000e+03	4726.00000	5080.000000	4829.000000	4780.000000	5080.000000
mean	1.235763e+05	3.21223	14.952362	16.509625	98.444561	14.790945
std	2.920847e+05	4.43865	9.033817	24.215410	58.164335	9.019765
min	-5.000000e+03	-1.00000	-5.000000	0.000000	-9.000000	-3.000000
25%	7.195629e+04	2.00000	7.000000	7.500000	48.000000	7.000000
50%	1.139949e+05	3.00000	15.000000	15.000000	97.000000	15.000000
75%	1.558942e+05	4.00000	23.000000	22.600000	149.000000	22.000000
max	9.999999e+06	99.00000	30.000000	481.911744	200.000000	30.000000

```
df.isnull().sum()
```

```
...      0
Employee_ID      0
First_Name       0
Last_Name        0
Gender           152
Date_of_Birth    201
Email            202
Phone            306
Department       0
Job_Title        0
Location         0
Date_of_Joining  0
Salary           251
Performance_Rating 354
Years_of_Experience 0
```

Human Resource Management and Employee Data Analysis

```
df.duplicated().sum()
```

```
np.int64(8)
```

STAGE 2 – Data Cleaning and Pre-processing

Handle missing values (impute or drop)

Handle duplicates

Treat outliers if required

Check skewness and apply transformations

Convert data types if needed

Feature transformations (date parts, derived fields if required for analysis)

```
print(df)
```

	Employee_ID	First_Name	Last_Name	Gender	Date_of_Birth	\
0	EMP2886	Sneha	Wang	Male	1995-12-01	
1	EMP2058	Yuki	Ahmed	Female	1984-10-09	
2	EMP5453	James	Ahmed	Other	06-24-1973	
3	EMP2412	Deepa	Johnson	Male	1998-12-18	
4	EMP2128	Fatima	Jones	Female	1977-05-07	
...	...	...	...	...	...	
5075	EMP2494	Chris	Wang	Male	1970-03-09	
5076	EMP4784	Karthik	kim	Male	1988-03-04	
5077	EMP3228	Omar	Singh	Male	1985-08-09	
5078	EMP4201	Mei	Smith	Male	1979-07-22	
5079	EMP1339	Ashley	Ahmed	Male	1985-02-18	

	Email	Phone	Department	\
0	NaN	NaN	Finance	
1	deepa.brown68@company.com	+91-6624030165	Admin	
2	layla.wang86@company.com	+91-6987938413	IT	
3	hiroshi.sharma54@company.com	+91-9379729728	Legal	
4	NaN	+91-8303574566	R&D	
...	...	...	...	
5075	omar.chen96@company.com	+91-7097328901	Admin	
5076	deepa.smith1@company.com	+91-6907212585	Customer Support	
5077	priya.williams48@company.com	+91-8731117944	Legal	
5078	rohan.martinez51@company.com	+91-7345953602	Fiance	
5079	michael.williams52@company.com	+91-9024190994	HR	

Human Resource Management and Employee Data Analysis

```
# Mode is used to fill missing values in categorical columns as it represents the most frequent and realistic category.
df.fillna({'Gender': df['Gender'].mode(), inplace=True)
```

```
# Mean is used to fill missing values in salary columns as it represents the average and maintains the overall distribution of the data.
df.fillna({'Salary': df['Salary'].mean(), inplace=True)
```

```
# Mode is used to fill missing values in Education_Level columns as it represents the most frequent and realistic category.
df.fillna({'Education_Level': df['Education_Level'].mode()[0], inplace=True)
```

```
# Mean is used to fill missing values in Performance_Rating columns as it represents the average and maintains the overall distribution of the data.
df.fillna({'Performance_Rating': df['Performance_Rating'].mean(), inplace=True)
```

```
# Missing values in Email and Phone were filled with "Not Available" to ensure data completeness and avoid null errors.
df['Email'] = df['Email'].fillna("Not Available")
```

```
df['Phone'] = df['Phone'].fillna("Not Available")
```

```
# Rows with missing Employee_ID were removed to ensure each record has a unique and valid identifier.
df.dropna(subset=['Employee_ID'], inplace=True)
```

```
# Date columns were converted to datetime format to ensure proper date handling and to handle invalid values as NaT.
df['Date_of_Birth'] = pd.to_datetime(df['Date_of_Birth'], errors='coerce')
```

```
df['Date_of_Joining'] = pd.to_datetime(df['Date_of_Joining'], errors='coerce')
```

```
df.drop_duplicates(inplace=True)
```

```
# Extra spaces were removed to ensure consistency in categorical data.
df['First_Name'] = df['First_Name'].str.strip()
df['Last_Name'] = df['Last_Name'].str.strip()
```

```
# Text was standardized to proper case to ensure consistency in categorical data.
df['Department'] = df['Department'].str.title()
df['Job_Title'] = df['Job_Title'].str.title()
```

```
# Keep only numeric values for consistency.
df['Phone'] = df['Phone'].str.replace(r'\D', '', regex=True)
```

```
▶ # Remove unrealistic salaries
df = df[df['Salary'] > 1000]
df = df[df['Salary'] < 1000000]
```

```
▶ # Select only numerical columns for finding how many outliers each column has.
num_cols = df.select_dtypes(include=['int64', 'float64']).columns

for col in num_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1

    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR
    outliers = df[(df[col] < lower_bound) | (df[col] > upper_bound)]
    print(f"{col}: {len(outliers)} outliers")
```

```
... Salary: 0 outliers
Performance_Rating: 21 outliers
Years_of_Experience: 0 outliers
Is_Active: 0 outliers
Bonus_Percentage: 25 outliers
Training_Hours: 0 outliers
Leaves_Taken: 0 outliers
```

Human Resource Management and Employee Data Analysis

```
# Boxplot visualization of the Performance_Rating column before removing outliers.
import matplotlib.pyplot as plt

df.boxplot(column='Performance_Rating')
plt.title("Before Removing Outliers")
plt.show()
```



```
# filter the outliers of Performance_Rating column
df = df[(df['Performance_Rating'] >= lower_bound) & (df['Performance_Rating'] <= upper_bound)]

print(f"Dataset after filtering outliers: {df.shape}")
```

... Dataset after filtering outliers: (5033, 20)

```
# Boxplot visualization of the Performance_Rating column After removing outliers.
df.boxplot(column='Performance_Rating')
plt.title("After Removing Outliers")
plt.show()
```



Human Resource Management and Employee Data Analysis

```
# Boxplot visualization of the Bonus_Percentage column before removing outliers.  
import matplotlib.pyplot as plt  
  
df.boxplot(column='Bonus_Percentage')  
plt.title("Before Removing Outliers")  
plt.show()
```



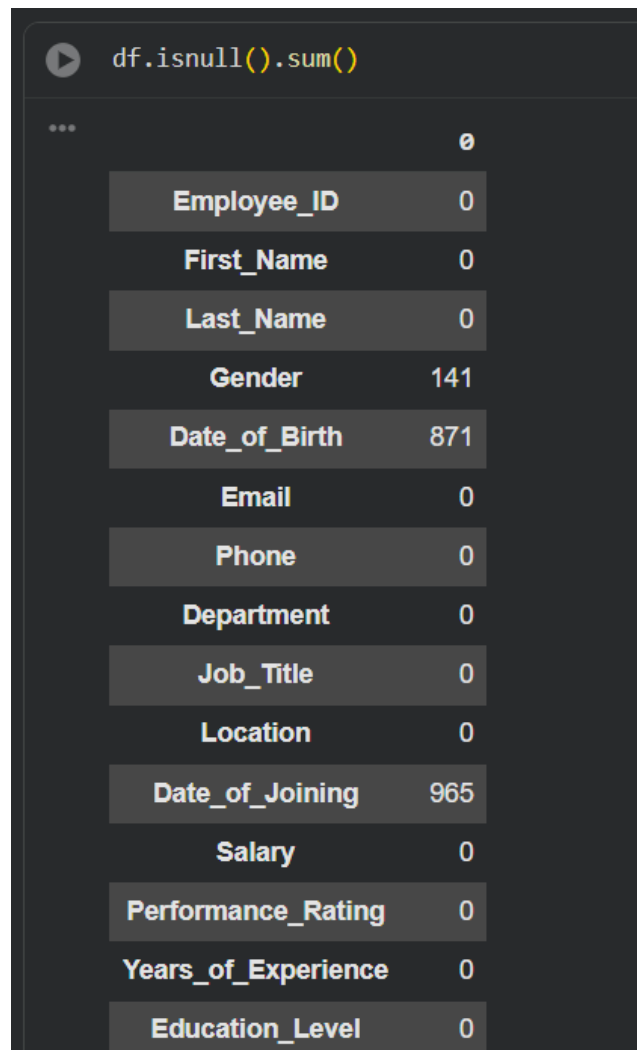
```
# filter the outliers of Bonus_Percentage column  
df = df[(df['Bonus_Percentage'] >= lower_bound) & (df['Bonus_Percentage'] <= upper_bound)]  
  
print(f"Dataset after flitering outliers: {df.shape}")
```

Dataset after flitering outliers: (4761, 20)

```
# Boxplot visualization of the Bonus_Percentage column After removing outliers.  
df.boxplot(column='Bonus_Percentage')  
plt.title("After Removing Outliers")  
plt.show()
```



Human Resource Management and Employee Data Analysis



A screenshot of a Jupyter Notebook cell showing the command `df.isnull().sum()` and its output. The output is a table with two columns: the column name and the count of null values. The counts are 0 for Employee_ID, First_Name, Last_Name, Email, Phone, Department, Job_Title, Location, Salary, Performance_Rating, Years_of_Experience, and Education_Level. The count is 141 for Gender and 871 for Date_of_Birth. The count is 965 for Date_of_Joining.

...	0
Employee_ID	0
First_Name	0
Last_Name	0
Gender	141
Date_of_Birth	871
Email	0
Phone	0
Department	0
Job_Title	0
Location	0
Date_of_Joining	965
Salary	0
Performance_Rating	0
Years_of_Experience	0
Education_Level	0

```
# Replaces missing values in the Gender column with "Unknown".
df['Gender'] = df['Gender'].fillna("Unknown")
```

```
# Converts the Date_of_Birth column to datetime format, setting invalid values to NaT.
df['Date_of_Birth'] = pd.to_datetime(df['Date_of_Birth'], errors='coerce')
```

```
# Converts the Date_of_Joining column to datetime format, setting invalid or incorrect values to NaT.
df['Date_of_Joining'] = pd.to_datetime(df['Date_of_Joining'], errors='coerce')
```

```
# Fills missing values in the Is_Active column with "No", assuming inactive status by default.
df['Is_Active'] = df['Is_Active'].fillna("No")
```

```
# Replaces missing values in the Bonus_Percentage column with the column's mean value.
df['Bonus_Percentage'] = df['Bonus_Percentage'].fillna(df['Bonus_Percentage'].mean())
```

```
# Replaces missing values in the **Training_Hours** column with the column's mean value.
df['Training_Hours'] = df['Training_Hours'].fillna(df['Training_Hours'].mean())
```

```
▶ # Fills missing values in the Manager_ID column with "No Manager", indicating no assigned manager.
df['Manager_ID'] = df['Manager_ID'].fillna("No Manager")
```

Human Resource Management and Employee Data Analysis

df.isnull().sum()

	0
Employee_ID	0
First_Name	0
Last_Name	0
Gender	0
Date_of_Birth	871
Email	0
Phone	0
Department	0
Job_Title	0
Location	0
Date_of_Joining	965
Salary	0
Performance_Rating	0

```
df['Date_of_Birth'] = df['Date_of_Birth'].fillna(df['Date_of_Birth'].median())
```

```
df['Date_of_Joining'] = df['Date_of_Joining'].fillna(df['Date_of_Joining'].median())
```

Human Resource Management and Employee Data Analysis

```
df.isnull().sum()
```

	0
Employee_ID	0
First_Name	0
Last_Name	0
Gender	0
Date_of_Birth	0
Email	0
Phone	0
Department	0
Job_Title	0
Location	0
Date_of_Joining	0
Salary	0
Performance_Rating	0
Years_of_Experience	0

```
df.to_excel("HR_Cleaned_Dataset.xlsx", index=False)
```

STAGE 3 – EDA and Visualizations

Univariate Analysis → distribution of single variables (countplot, histogram, boxplot)

Bivariate Analysis → relation between two variables (scatterplot, barplot, correlation heatmap)

Multivariate Analysis → relation among 3+ variables (pairplot, grouped analysis, pivot tables, advanced plots)

Interpretation MUST with every visualization

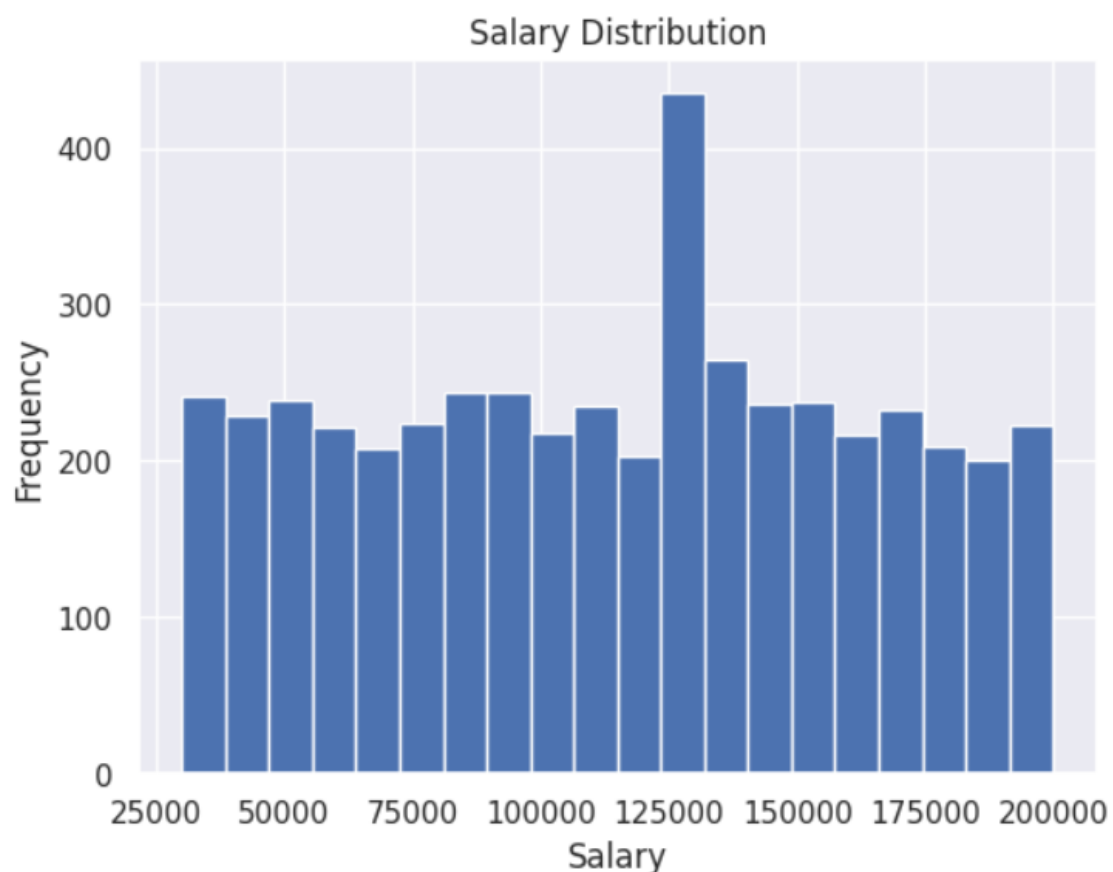
Focus on business story not just charts

Each chart must be in separate cells.

```
# Import Libraries
import matplotlib.pyplot as plt
import seaborn as sns
sns.set()
```

Human Resource Management and Employee Data Analysis

```
# Salary Distribution
plt.figure()
plt.hist(df['Salary'], bins=20)
plt.title("Salary Distribution")
plt.xlabel("Salary")
plt.ylabel("Frequency")
plt.show()
```

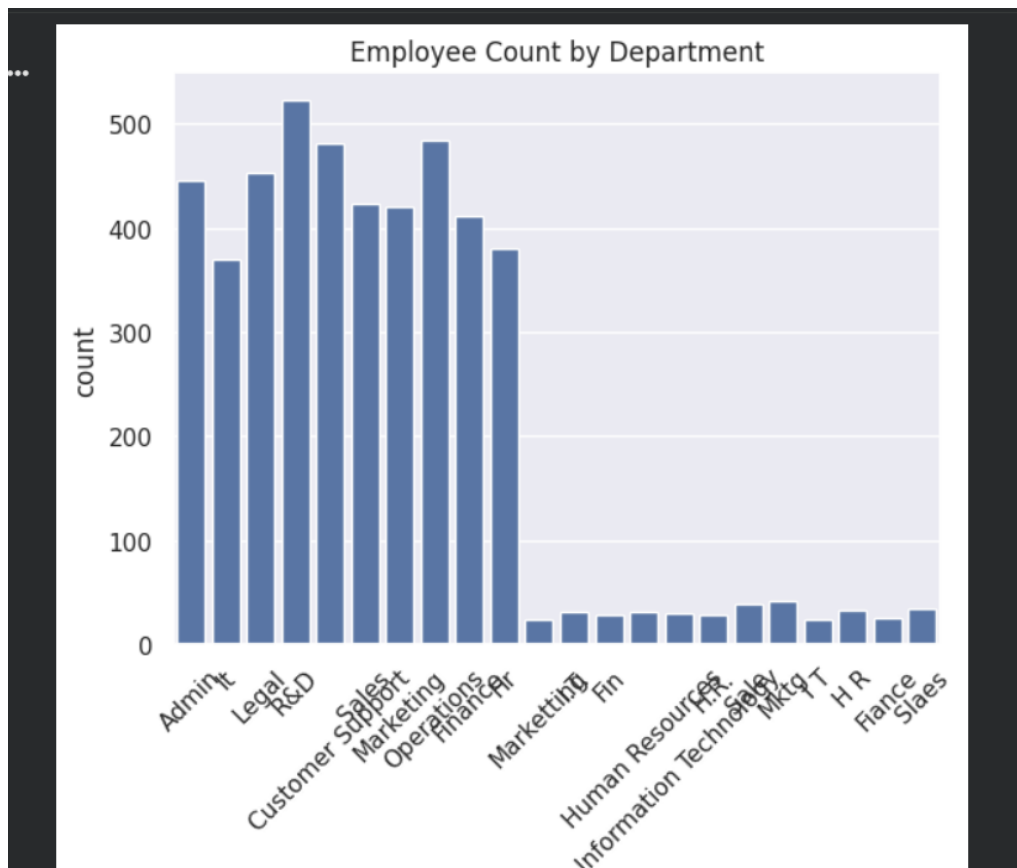


Interpretation of Histogram chart:

- The histogram represents the distribution of employee salaries in the dataset. It shows that most employees fall within a mid-range salary bracket, while only a few employees earn very high or very low salaries.
- This indicates that the salary structure is somewhat concentrated around the average, with limited extreme values.
- By using the Salary feature, the chart helps in understanding how salaries are spread across the organization and highlights any imbalance or outliers.
- This insight is useful for HR in evaluating compensation policies and ensuring a fair and consistent pay structure.

Human Resource Management and Employee Data Analysis

```
# Department Count
plt.figure()
sns.countplot(x='Department', data=df)
plt.xticks(rotation=45)
plt.title("Employee Count by Department")
plt.show()
```

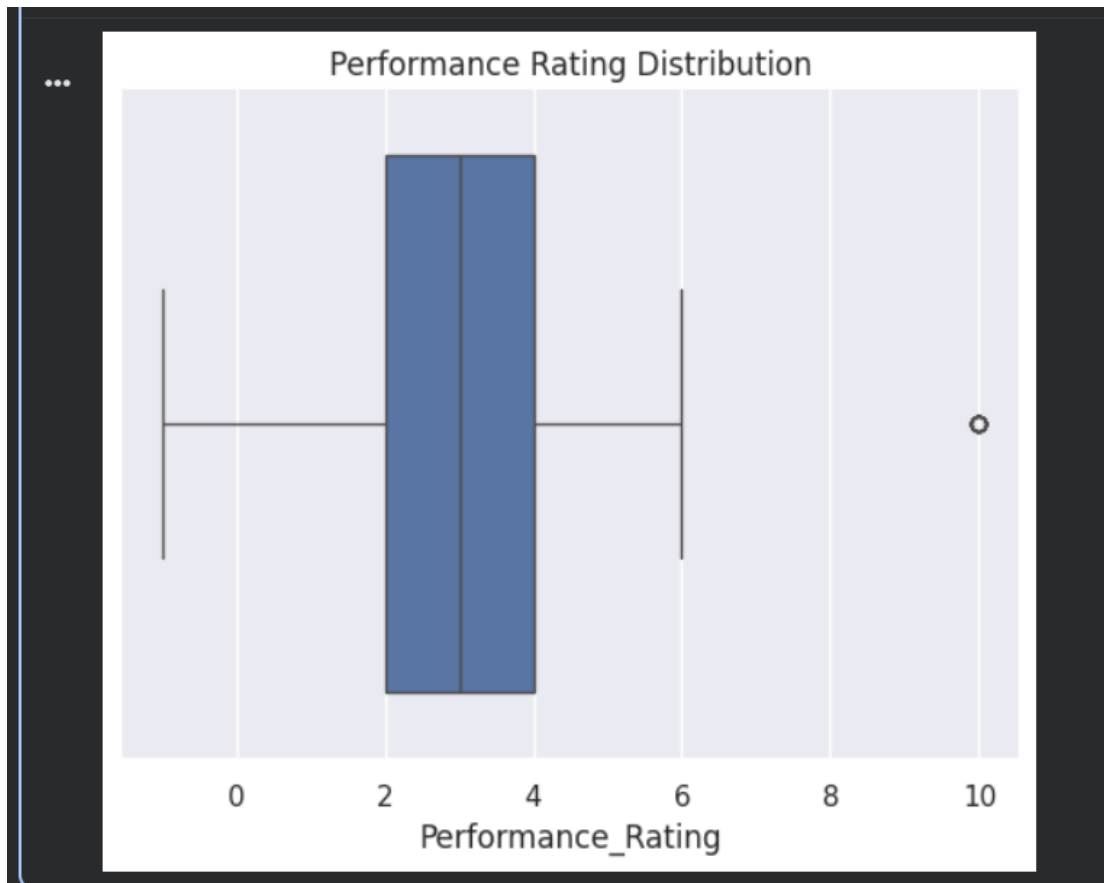


Interpretation of Countplot chart

- The countplot represents the number of employees in each department using the Department feature.
- It shows that some departments have a higher number of employees while others have fewer, indicating an uneven distribution of the workforce across the organization.
- This suggests that certain departments may require more resources or have higher operational demand compared to others.
- From this chart, we are showing how employees are distributed across departments, which helps in understanding workforce allocation and supports HR decisions related to hiring, resource planning, and organizational structure.

Human Resource Management and Employee Data Analysis

```
# Performance Rating
plt.figure()
sns.boxplot(x=df['Performance_Rating'])
plt.title("Performance Rating Distribution")
plt.show()
```

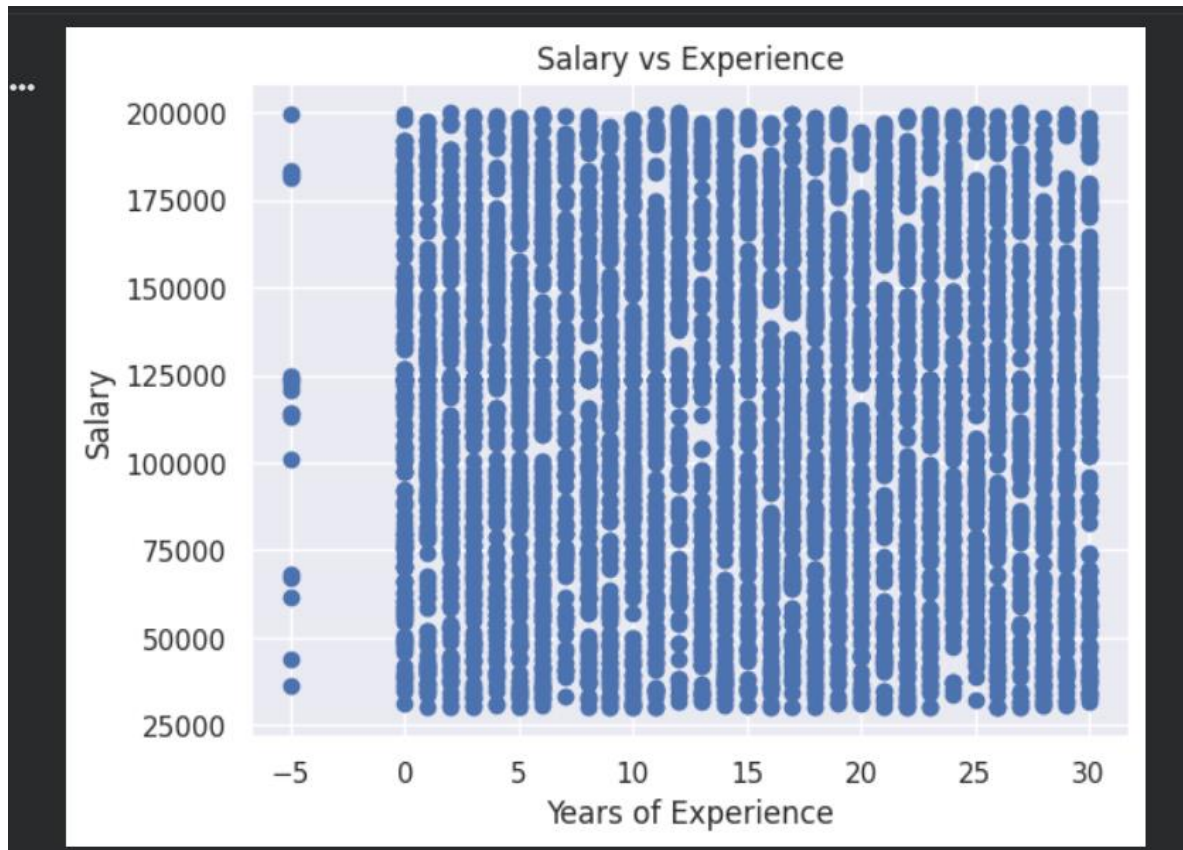


Interpretation of Boxplot chart

- The boxplot represents the distribution of employee performance ratings using the Performance_Rating feature.
- It shows the median rating, the spread of most values, and any extreme values (outliers).
- The chart indicates that most employees have performance ratings within a consistent range, suggesting stable overall performance across the organization, while a few outliers represent exceptionally high or low performers.
- From this chart, we are showing how performance ratings are distributed among employees, which helps HR identify general performance trends and detect employees who may need recognition or improvement.

Human Resource Management and Employee Data Analysis

```
# Salary vs Experience
plt.figure()
plt.scatter(df['Years_of_Experience'], df['Salary'])
plt.xlabel("Years of Experience")
plt.ylabel("Salary")
plt.title("Salary vs Experience")
plt.show()
```

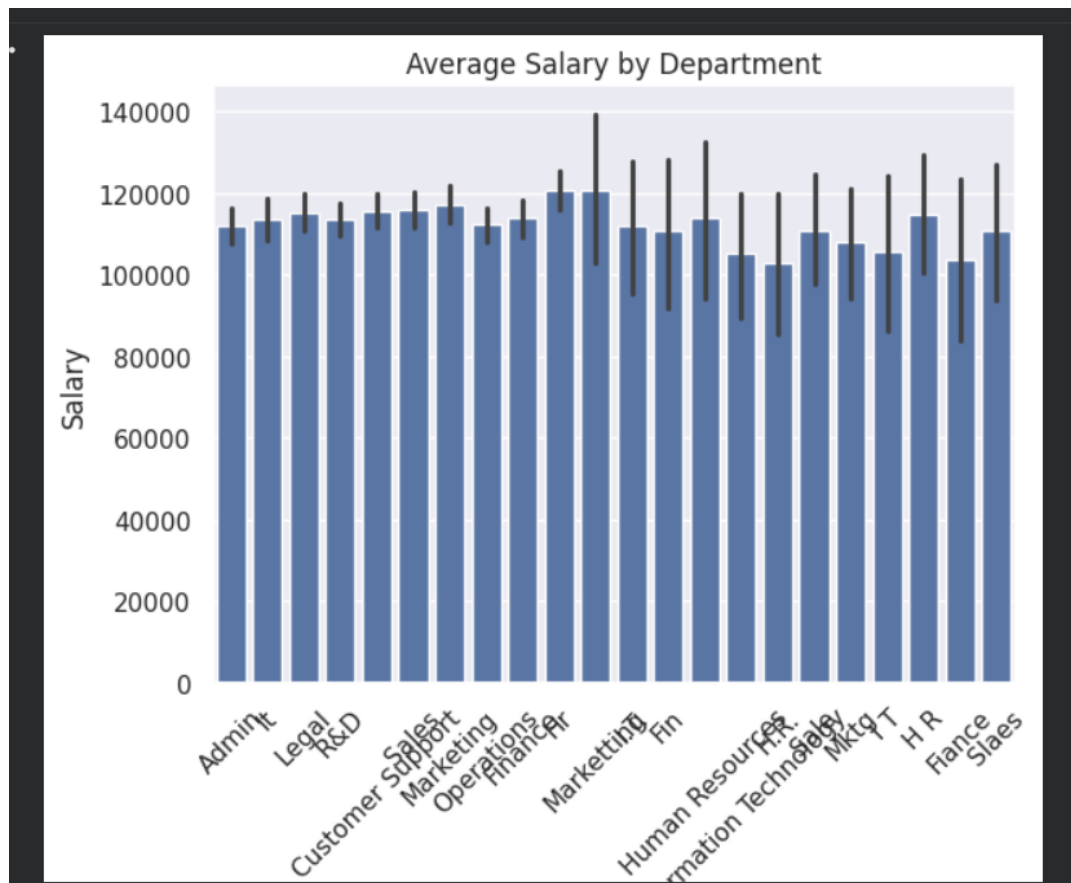


Interpretation of Scatterplot chart

- The scatterplot shows the relationship between employees' years of experience and their salary using the Years_of_Experience and Salary features.
- It indicates a general positive trend, where employees with more experience tend to earn higher salaries, although there are some variations.
- This suggests that experience plays a key role in determining compensation, but other factors may also influence salary.
- From this chart, we are showing how salary grows with experience, which helps HR evaluate whether the organization follows a fair and consistent pay structure based on experience.

Human Resource Management and Employee Data Analysis

```
# Salary by Department
plt.figure()
sns.barplot(x='Department', y='Salary', data=df)
plt.xticks(rotation=45)
plt.title("Average Salary by Department")
plt.show()
```

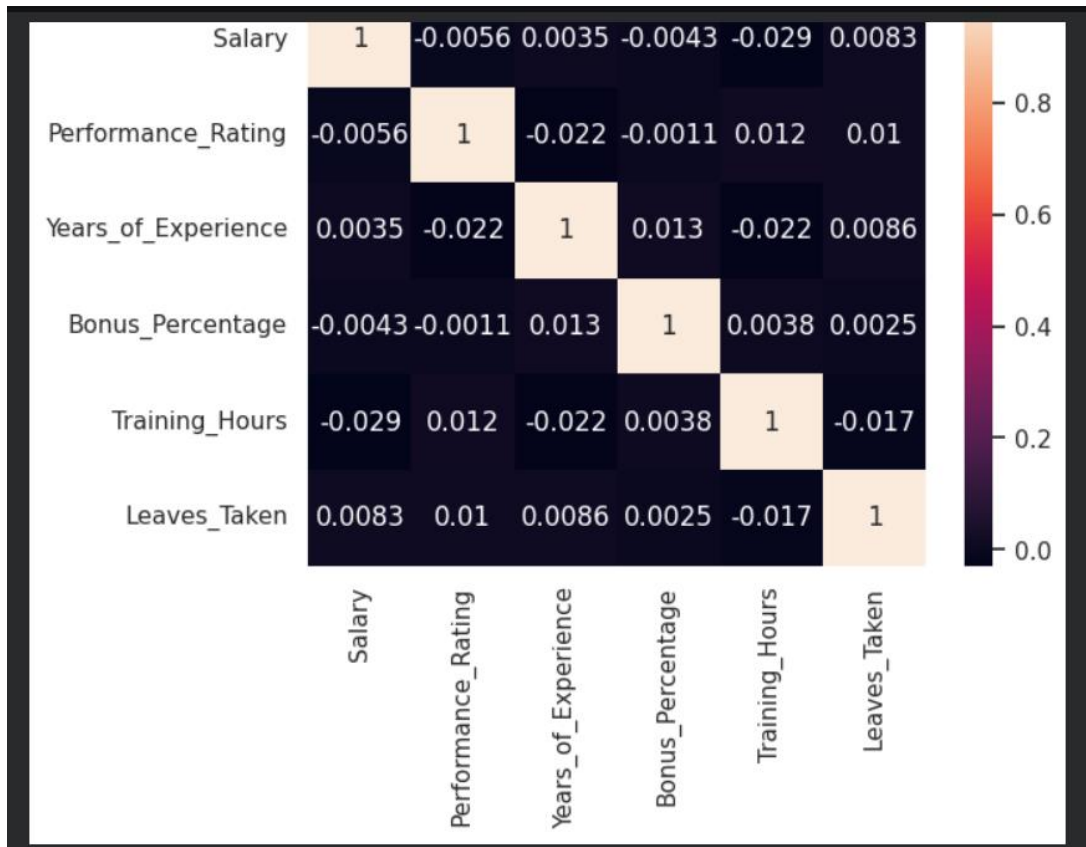


Interpretation of Barplot chart

- The barplot shows the average salary across different departments using the Department and Salary features.
- It indicates that some departments have higher average salaries while others have comparatively lower salaries, reflecting differences in job roles, skills, and responsibilities.
- This suggests that compensation varies based on the nature and demand of work in each department.
- From this chart, we are showing a comparison of department-wise salary levels, which helps HR understand pay distribution, manage budgets, and design fair compensation strategies across the organization.

Human Resource Management and Employee Data Analysis

```
# Correlation Heatmap
plt.figure()
sns.heatmap(df.corr(numeric_only=True), annot=True)
plt.title("Correlation Heatmap")
plt.show()
```

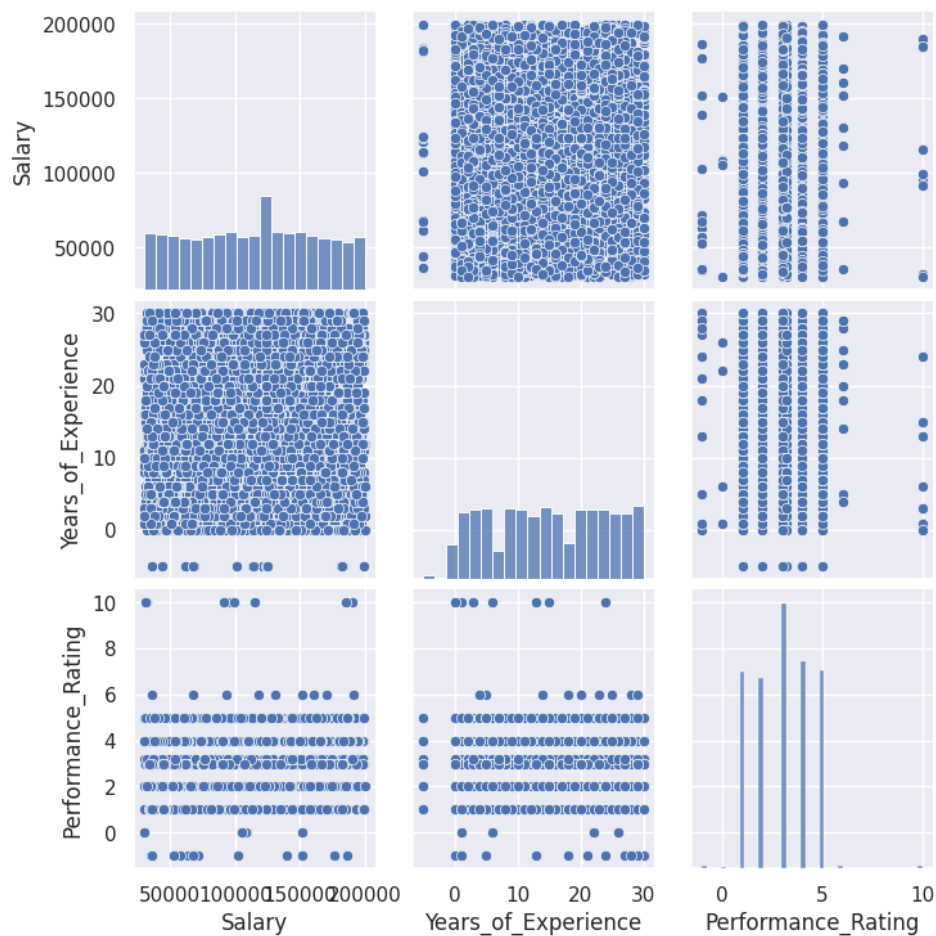


Interpretation of Correlation Heatmap chart

- The correlation heatmap visualizes the relationships between numerical features in the dataset such as Salary, Years_of_Experience, Performance_Rating, Bonus_Percentage, and Training_Hours.
- Each cell shows the correlation value between two variables, where values closer to +1 indicate a strong positive relationship and values closer to -1 indicate a negative relationship.
- The chart suggests that certain variables, such as experience and salary, may have a positive correlation, meaning that as experience increases, salary tends to increase as well.
- From this chart, we are showing how different numerical factors are related to each other, which helps identify key drivers influencing salary and performance and supports data-driven HR decision-making.

```
# salary relates to experience and performance
sns.pairplot(df[['Salary', 'Years_of_Experience', 'Performance_Rating']])
plt.show()
```

Human Resource Management and Employee Data Analysis



Interpretation of Pairplot chart

- The pairplot visualizes the relationships between multiple numerical features—Salary, Years_of_Experience, and Performance_Rating—by showing scatterplots for each pair and distributions along the diagonal.
- It indicates that salary generally increases with experience, while the relationship between performance rating and salary or experience appears weaker or less consistent.
- The diagonal plots show how each variable is distributed individually.
- From this chart, we are showing how these key factors interact with each other, helping to identify patterns, trends, and potential influences on employee compensation and performance, which is useful for HR decision-making and strategy.



```
# Pivot Table (Grouped Analysis)
pivot = df.pivot_table(values='Salary', index='Department', columns='Education_Level', aggfunc='mean')
pivot
```

Human Resource Management and Employee Data Analysis

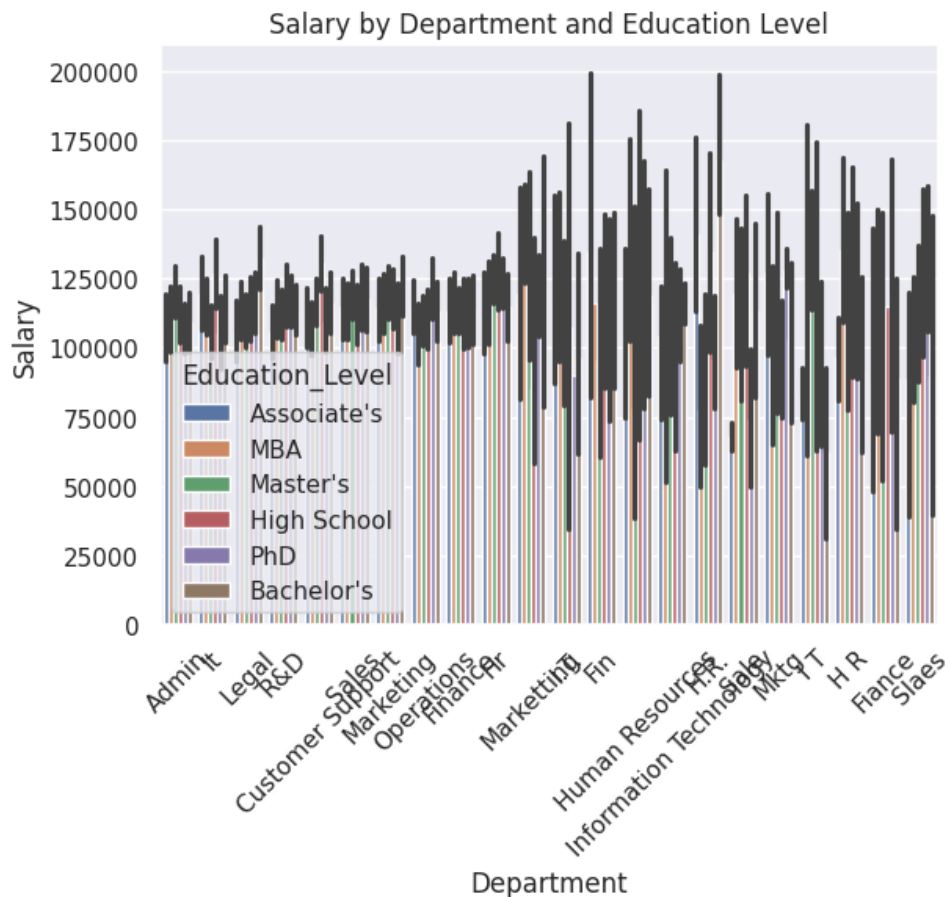
Education_Level	Associate's	Bachelor's	High School	MBA	Master's	PhD
Department						
Admin	106892.664524	109489.402720	111470.721423	110490.721285	120841.886546	106948.248189
Customer Support	110519.993536	115828.408669	130594.049323	107064.058731	116992.496281	109513.484091
Fiance	89656.467500	88667.310000	114686.470000	107520.068571	99756.470000	121193.896000
Fin	135021.074697	125768.706667	114713.775000	115840.000000	95642.666250	107192.202013
Finance	113268.542311	114380.339505	112372.651550	116215.866123	113678.360986	112952.873704
H R	96626.792000	91991.282500	125842.166818	135524.573333	111598.610584	124330.062222
H.R.	141059.030000	167464.203333	134622.530000	77342.707778	87348.591250	96268.776667
Hr	112312.535594	115068.886133	127776.172169	116321.812611	125052.385776	123282.124760
Human Resources	105120.682857	119074.365000	126236.255000	132247.057500	77926.852500	125299.690000
I T	82776.550000	62098.000000	118500.650000	121853.624545	134460.008000	94139.364000
I.T.	117547.084818	96072.288333	107796.645000	128157.878571	110288.087000	89761.270000
Information Technology	104344.499697	116311.144697	100640.120818	91860.246667	107118.701281	111835.725000
It	119264.670325	114063.186429	127070.263785	115058.854917	105426.549907	105516.656002
Legal	106500.654766	132412.106711	114324.546417	113669.733781	110231.400490	116786.571271
Marketing	114167.908571	122689.269016	117726.684242	115630.583801	120193.144293	110578.076882
Marketting	119623.732818	121286.610000	85430.170000	141335.897045	130475.665714	119081.835000

Interpretation of Pivot Table (Grouped Analysis)

- The pivot table summarizes the average salary across different departments and education levels using the Salary, Department, and Education_Level features.
- It shows how salary varies not just by department but also by the qualification of employees within each department.
- The table indicates that employees with higher education levels generally earn higher salaries, although this pattern may differ across departments.
- From this analysis, we are showing the combined impact of department and education on salary, which helps HR understand compensation structure, plan salary policies, and make informed decisions regarding hiring and promotions.

Human Resource Management and Employee Data Analysis

```
# Salary by Department & Education
plt.figure()
sns.barplot(x='Department', y='Salary', hue='Education_Level', data=df)
plt.xticks(rotation=45)
plt.title("Salary by Department and Education Level")
plt.show()
```



Interpretation of Barplot chart

- The barplot compares average salaries across departments while also distinguishing employees by their education level using the Department, Salary, and Education_Level features.
- It shows that salary varies not only between departments but also within each department based on education level, where employees with higher qualifications generally earn more.
- However, the impact of education on salary may differ from one department to another.
- From this chart, we are showing the combined effect of department and education on compensation, which helps HR understand pay structure, evaluate fairness, and design better salary and promotion policies.

Human Resource Management and Employee Data Analysis

STAGE 4 – Documentation, Insights and Presentation

Show a dashboard aligning all the charts in powerbi by connecting python in PowerBI

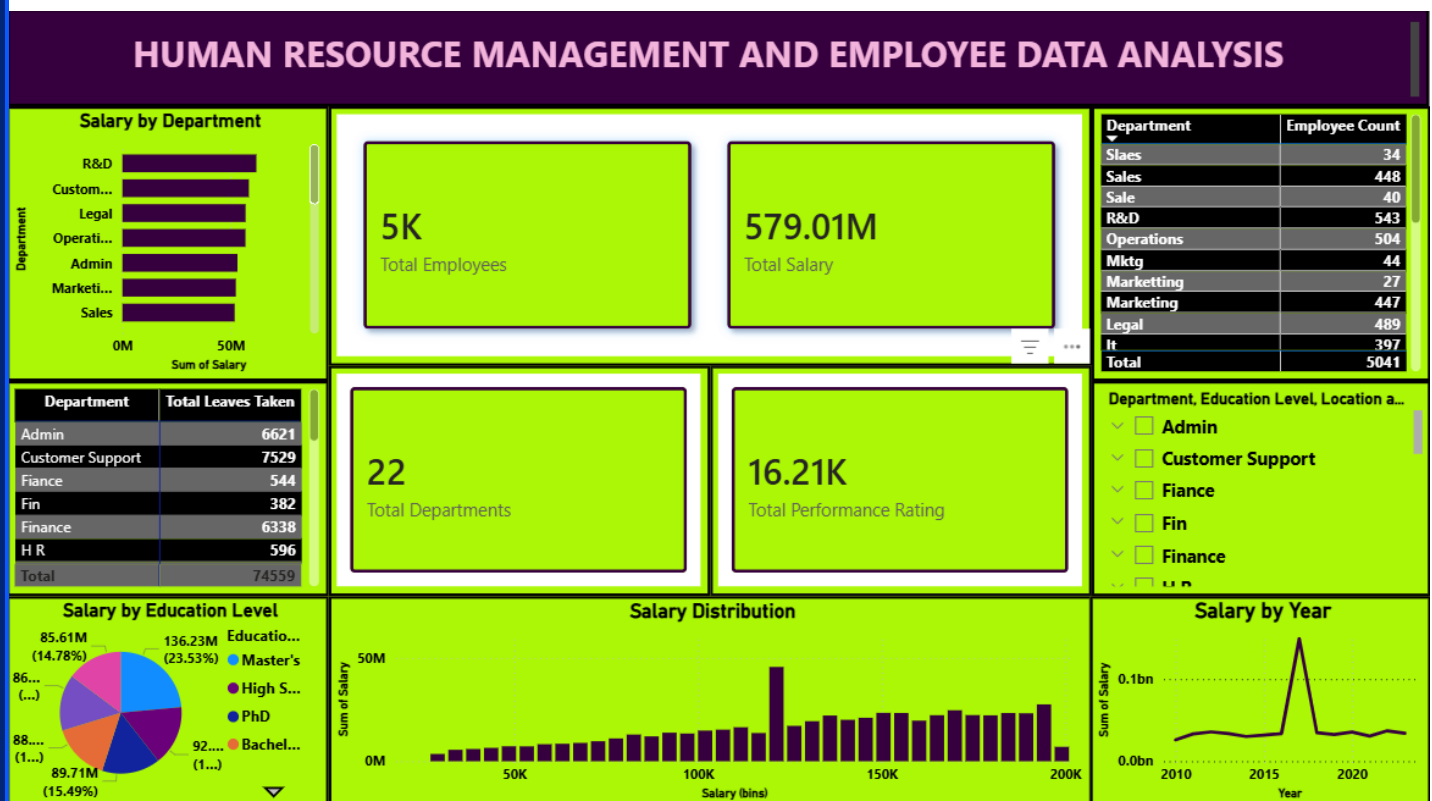
Summarize findings in plain English (what do the patterns mean)

Highlight 3–5 key insights (trends, anomalies, correlations)

Provide recommendations for business or decision-making

Explain the FINAL STORY WITH THE DASHBOARD

Create a PDF pasting your DASHBOARD, explaining all the above mentioned concepts and submit it



The dashboard shows that the organization has around **5,000 employees** with a total salary expense of **579M**, indicating a large workforce and high compensation cost. Salaries are not evenly distributed—most employees fall within a mid-range salary, while a few earn significantly higher amounts. Different departments show variations in both employee count and salary, suggesting differences in roles and responsibilities. Education level also impacts salary, where higher qualifications tend to receive better pay. Overall, salary is influenced by multiple factors like experience, department, and education.

Human Resource Management and Employee Data Analysis

KEY INSIGHTS

◆ Insight 1: Salary Concentration

Most employees earn in the mid salary range, with fewer high earners → indicates a structured pay system with limited extremes.

◆ Insight 2: Department Differences

Some departments (like R&D, Finance, Marketing) have higher salary totals, showing they are cost-intensive or high-value roles.

◆ Insight 3: Workforce Distribution

Employee count is uneven across departments → some departments are heavily staffed while others are small.

◆ Insight 4: Salary Spike (Anomaly)

There is a noticeable salary spike in a specific year (around 2017) → could indicate bonuses, hiring surge, or policy change.

◆ Insight 5: Education Impact

Higher education levels (Masters, PhD) are associated with higher salary contribution, showing qualification-based pay structure

BUSINESS RECOMMENDATIONS:

1. Balance Salary Structure

Review departments with very high salary costs and ensure fair compensation distribution.

2. Optimize Workforce Allocation

Departments with too many employees should be evaluated for efficiency and productivity.

3. Monitor Salary Spikes

Investigate sudden increases (like 2017 spike) to ensure budget control and consistency.

4. Promote Skill-Based Growth

Encourage employees to upgrade education/skills since it clearly impacts salary growth.

5. Data-Driven HR Decisions

Use dashboard insights regularly for hiring, promotions, and salary planning.

FINAL STORY WITH DASHBOARD

The dashboard provides a complete overview of the organization's workforce and salary structure. It tells a story where salary is not random but influenced by experience, department, and education level. While the company maintains a stable workforce, there are clear differences in how employees are distributed and compensated across departments. Some areas demand higher salaries due to their importance, while others are more resource heavy. The presence of salary spikes indicates periods of change in the organization. Overall, the dashboard helps HR and management make informed decisions about compensation, hiring, and resource allocation, ensuring efficiency and fairness within the company.

CONCLUSION

The analysis highlights that salary is influenced by experience, department, and education, while workforce distribution varies across departments. These insights help improve HR decision-making and organizational efficiency.